ATTACHMENT -C

EMPLOYER

WEST AFRICAN ENERGY SA

SENEGAL NATIONAL ELECTRICITY AGENCY

EPC CONTRACTOR

ÇALIK ENERJİ SAN. VE TİC. A.Ş

CAP DES BICHES COMBINED CYCLE POWER STATION PROJECT (300 MW)

REVISION HISTORY

REV.	DATE	DESCRIPTION	PREPARED	CHECKED	APPROVED
F	23.08.2021	FOR REVIEW	CSR	TLM	SSR
E	23.07.2021	FOR REVIEW	MV	TLM	SSR
D	05.07.2021	FOR REVIEW	MV	TLM	SSR
C	04.03.2021	FOR REVIEW	MV	TLM	SSR
B	21.01.2021	FOR REVIEW	MV	TLM	SSR
A	12.01.2021	FOR REVIEW	MV	TLM	SSR

DOCUMENT TITLE/SUBJECT:

GENERAL DESIGN CRITERIA

PROJECT DOCUMENT NO:

Project	*Unit*	*System*	*Disc*	*Doc.*	*Type*	*Orig.*	*Serial No*	Detailed Information	**Rev.**	**F**
SNC	**36**	**VD**	**IR**			**CEI**	**001**		**Size**	**A-4**
									Lang.	**Eng.**
									Total Sheet	**21**

VENDOR DOCUMENT NO: **** if applicable***

INDEX:

1. INTRODUCTION

1.1. PURPOSE

The purpose of the document is to furnish the general design criteria for the plant.

1.2. RELEVANT PARTIES

West African Energy	:	Employer
CALIK ENERJi”	:	Contractor
Fichtner India	:	Engineering Sub Contractor
Supplier	:	Equipment Supplier

2. PROJECT DESCRIPTION

2.1. GENERAL

West African Energy intends to build a combined cycle power plant at Dakar.CALIK ENERJi” and has received the order from EMPLOYER to execute the power plant on an Engineering, Procurement, Construction and Commissioning (EPC) basis.

2.2. PROJECT LOCATION

Cap des Biches site is on the edge of the Atlantic Ocean, located in the North Industrial Zone of Dakar, 25 KM from the port of Dakar.

Unloading of equipments and imported materials will be made from the port of Autonomous port of Dakar.

2.3. PLANT CONFIGURATION

The plant configuration comprise of One (1) combined cycle power block in multi shaft configuration. The block consists of two (2) gas turbine and generators (GTGs), two (2) heat recovery steam generators (HRSGs) & one (1) steam turbine and generator (STG), and Balance of Plant (BOP) equipment. The gas turbine is capable of firing trifuels - Natural gas (Primary fuel), Naptha (secondary fuel) and Gas Oil (startup fuel) respectively.

The gas turbine will be of Frame 9E.03. The HRSG will be of triple pressure, non reheat and unfired type. The Steam turbine will be of axial exhaust type with air cooled condenser. The gas turbine, HRSG and steam turbine are supplied by GE.

2.4. SYSTEM AND EQUIPMENT NUMBERING

The plant system and equipment numbering will follow KKS numbering.

2.5. PLANT COORDINATE SYSTEM

The plant UTM coordinates are 28 P, 252268.92 E, 1628766.44 N

2.6. DESIGN LIFE

Design life shall be to operate for a minimum of 25 years.

2.7. OPERATING MODE AND CONTROL PHILOSOPHY

Operating Mode	Base Loaded, Thirty (30) Startups per Year, per GT
Simple Cycle Operation	Yes, Bypass stack provided
Plant Control Philosophy	Remote Manual Startup/Shutdown; Automatic Operation Under load from control room
Remote Dispatch Capability	Included, from control room

2.8. REDUNDANCY

The design shall ensure a minimum requirement of manual intervention in case of failure of an individual plant component or system failure. In the event of a component or subsystem failure, automatic recovery of the redundant equipment or subsystem must be automatic. Equipment failure must not prevent the continuous operation of the plant in automatic mode.

To achieve this, the Supplier is required to provide the necessary redundancy of the Plant, equipment controls and protection whether specified herein or not. This redundancy concept will become a matter of approval by Contractor, Employer and Engineering Sub contractor.

Components or sections of the system that are essential to the plant availability are of redundant design.

2.9. ENVIRONMENTAL DATA

S.No	Description	Data	
A.	Altitude	20 m + NGA approximately.	
B	Ambient Temperature	Design	26° C
		Max	41° C
		Min	10 °C
C	Relative humidity	Design	70%
		Max	98%
		Min	32%

Seismicity

For seismicity, the following value shall be considered:

The Dakar region is classified ZONE 0 in accordance with UBC 97 Soil Profile type SD and Occupancy Category I.

Wind

All exposed structures or parts of structures shall be designed to withstand wind pressures in all directions, in accordance with Eurocode 1

Wind direction : North and North-West

The ambient is characterized by a salty air environment with dust concentration

The basic wind speed	:	161km / hr
Annual Rain fall	:	2500 mm
Ground Snow Load	:	0 kg/cm2

3. CODES AND STANDARDS

3.1. GENERAL

The supplier shall be responsible for the basic and detailed design, the choice and sizing of components, the selection of materials, the manufacturing, the testing at fabricator's works, the erection and commissioning of every component of the Work and of the Plant as a whole, so that the Work and the Plant as a whole and every such component shall strictly comply with the requirements defined by the Senegal statutory regulations, Prudent Utility Practices and with internationally recognized standards, guides, codes and practices.

The reference list included in the following is not intended to be complete or limiting and in particular any applicable statutory regulations, even if not included in the list, shall be strictly complied with.

Any conflicts and discrepancies between the applicable standards shall be resolved according to the following priority of the applicable documents:

- Statutory Regulations
- Technical Specification
- Industrial Codes and Standards

If more than one of the Standards included in the following list shall apply to a specific design aspect, the Contractor shall require to the Employer an indication of the standard to be applied. All the codes and Standards shall be applied in the revision current at the time of signing of this contract. F

Civil Works and Structures

Structural works

F

EN 1990 Eurocode	:	Basis of structural design
EN 1991 Eurocode 1	:	Actions on structures
EN 1993 Eurocode 3	:	Design of steel structures
EN 10025	:	Structural steel standards
EN 1998 Part-6	:	Towers, masts and chimneys
EN 10025	:	Structural steel standards
EN 1993-1-1	:	General rules for buildings
EN 1993-1-5	:	Plated structural elements
EN 1993-1-8	:	Design of joints
EN 1993-1-10	:	Material toughness and through-thickness Properties

Civil works

EN 1992 Eurocode 2	:	Design of concrete structures
EN 1992 Part1-6	:	Plain concrete structures
EN 1992 Part1-3	:	Liquid retaining and containing structures
EN 1993-1-5	:	Piling
EN 1996 Eurocode 6	:	Design of Masonry structures
EN 1997 Eurocode 7	:	Geotechnical design
EN 1998 Part-5	:	Foundations and geotechnical aspects
UBC-1997	:	Uniform Building Code (for Seismic)
DIN 4024, CP2012, ACI 351.3R-04	:	Foundations for Dynamic Equipment
DIN EN	:	German Institute for Standardization
ANSI/ASCE 7.88	:	Minimum Design Loads for Buildings and Other Structures
ACI	:	American Concrete Institute
AISC	:	Design Standards for Metallic Framed Structures

AWS	:	Design standards for Structure welding
ANSI/ASCE 7.93	:	Structural Load Design of Buildings

ASTM International

DTU	:	Reinforced Concrete design
ASCE	:	American Society of Civil Engineers
AISC	:	American Institute of Steel Construction
AWS	:	American Welding Society
IBC 2006	:	International Building Code
UFC 2013	:	Unified Facilities Criteria Issue 2013
NFPA	:	National Fire Protection Association
BS	:	British Standards

Local Laws and Building Codes

Note: The code hierarchy shall be as follows: F

1. Eurocode,
2. American code,
3. British code and
4. Local code

Mechanical Plants

Boiler and pressure vessels

ASME sec I "Power boilers"

ASME sec VIII "Pressure vessels"

PED Pressure Equipment Directive

Steam Turbine

IEC 45 Steam Turbines F

ISO 20816-1 Mechanical vibration - Measurement and evaluation of machine vibration - Part 1: General guidelines

ISO 20816-2 Mechanical vibration - Measurement and evaluation of machine vibration - Part 2

Materials

DIN Deutsches Institut für Normung

ASME sec II "Materials" F

ASTM American Society for Testing and Materials

Pumps

DIN Deutsches Institut für Normung

ASME American Society of Mechanical Engineers

HI Hydraulic Institute

Heat Exchangers

TEMA Tubular Exchangers Manufacturers Association (class "C")

HEI Heat Exchangers Institute

NFPA 20 Standard for the Installation of Stationary Pumps for Fire Protection

ISO 5199 Technical specifications for centrifugal pumps

ISO 9906 Rotodynamic pumps — Hydraulic performance acceptance tests — Grades 1, 2 and 3

ANSI/HI 14.3 - Rotodynamic Pumps for Design and Application

ANSI/HI 14.6 - Rotodynamic Pumps for Hydraulic Performance Acceptance Tests

F

Tanks

AWWA Steel tanks for water storage

API Tanks for Fuel

Condenser and air extraction system

HEI Heat Exchangers Institute

Gearbox

AGMA American Gear Manufacturer Association

Valves

ANSI B16.34 "Steel butt-welding end valves"

ANSI B16.10 "Face to face and end to end dimensions of ferrous valves"

MSS-SP25 "Standard marking systems for valves, fittings, flanges and unions"

MSS-SP45 "By-pass and drain connection standards"

MSS-SP72 "Ball valves with flanged or butt welding ends for general service"

MSS-SP70 "Cast iron gate valves flanged and threaded ends"

MSS-SP71 "Cast iron swing check valves, flanged and threaded ends"

MSS-SP85 "Cast iron globe and angle valves flanged and threaded ends"

MSS-SP80 "Bronze gate, globe, angle and check valves"

MSS- SP84 "Steel valves- socket welding and threaded ends"

MSS-SP88 "Diaphragm type valves"

NFE "Mechanical Norms for Hollow Sections and Valves"

AWWA C500 "Gate valves for ordinary water works service"

Piping

ANSI B31.1 "Power piping"

ANSI B36.10 "Welded and seamless wrought steel pipe"

ANSI B36.19 "Stainless steel pipe"

ANSI B16.34 "Valves, Flanges, Fittings and Gaskets" F

ANSI B16.5 "Steel pipe flanges and flanged fittings"

ANSI B16.47 " Large Diameter Steel Flanges – NPS – 26" to 60"."

AWWA C207 "Steel pipe flanges for waterworks service"

MSS-SP44 "Steel pipe line flanges"

ANSI B16.20 "Ring-joint gaskets and grooves for steel pipe flanges"

ANSI B16.21 "Non Metallic Gaskets for Pipe Flanges"

ANSI B18.2.1 "Square and ex bolts and screws

ANSI B18.2.2 "Square and ex nuts"

ANSI B1.1 "Unified inch screw threads"

ANSI B2.1 "Pipe threads"

ANSI B16.9 "Factory-made wrought steel butt-welding fittings"

ANSI B16.11 "Forged steel fittings socket welding and threaded"

ANSI B16.25 "Butt-welding ends"

ANSI B16.28 "Wrought steel butt welding short radius elbows and returns" F

Welding and non destructive testing

ASME V " Non destructive Examination" F

ASME IX "Welding and brazing qualification"

AWS American Welding Society F

WRC 107/ WRC 297 Local Stresses In Cylindrical Shells Due To External Loadings on Nozzles

DIN 8560: Welders Qualification

MSS SP 58 "Pipe Hangers and Supports - Materials, Design, Manufacture, Selection, Application, and Installation"

MSS SP 69 "Pipe Hangers and Supports - Selection and Application". F

NFA" Metallurgical Norms for Sheetings and Hollow Sections"

NFEN 287" Welders Qualification"

NFEN 288 "Welding Procedures"

Painting

SSPC-SP3: Society of Protective Coatings (SSPC) – Power Tool Cleaning

SSPC-SP6: Society of Protective Coatings (SSPC) – Commercial Blast Cleaning

SSPC_SP10: Society of Protective Coatings (SSPC) – Near-White Blast Cleaning

ISO 12944 - Corrosion protection of steel structures by protective paint systems

BS EN ISO 4618 - Paints and varnishes - Terms and definitions

Hoists and cranes

EN 15011 Cranes, Bridge and portal cranes

FEM – European Federation of Material Handling and Packaging

Heat, ventilation and air conditioning

ASHRAE "Fundamentals 1997"

SMACNA - Sheet Metal and Air Conditioning Contractors' National Association

ASHRAE 15 - Safety Standard for Refrigeration Systems

ASHRAE 62.1 - Ventilation for Acceptable Indoor Air Quality

ASHRAE 55 - Thermal environmental conditions for human occupancy

AHRI - Air Conditioning, Heating, and Refrigeration Institute

ANSI/HI - American National Standard for Rotodynamic pumps, Hydraulic Institute, U.S.A

Vessels

BS 5500 "Pressure Vessels Specifications"

CODAP "Pressure Vessels Fabrication"

TRD 301 "Cylindrical Pressurised Vessels"

TRD 303 "Spherical Pressurised Vessels"

NFE 29.005 "Pressurised pipework"

DIN 2413 "Steel Pipes" F

Electrical System

Electrical generator

IEC ISO-1 "Rotating electrical machines"

IEC 60034 “Rotating electrical machines”

NFC 51.111 “Rotating electrical machines”

IEC 60085 "Thermal evaluation and classification of electrical insulation"

IEEE Std. 122 "Functional and performance characteristics of control systems for steam turbine generator units”

ANSI/IEEE 421 “Standard definitions for excitation systems for synchronous machines”

IEC 60694 “Generator switchgear”

ISO 8528-1 “Reciprocating internal combustion engine driven alternating current generating sets”

Transformers

IEC 60726 “Dry-type power transformers”

IEC 60076 "Power transformers"

IEC 60296 "Isolating mineral oils"

IEC 60146 “Semiconductor converters”

IEC 61378 “Converter transformers”

IEC 60551 “Determination of transformer and reactor sound level”

IEC 61869-2 “Instrument Transformers – Part 3: Additional requirements for Current Transformers”

IEC 61869-3 “Instrument transformers - Part 3: Additional requirements for Inductive Voltage Transformers”

Switchgears and control gears

IEC 62271 “High-Voltage switchgear and control gear”

IEC 60376 “Specification and acceptance of new sulphur hexafluoride”

IEC 60480 “Guide to checking sulphur hexafluroide (SF6)

IEC 60529 “Degrees of protection provided by enclosures”(IP code)

IEC 60298 “HV Metal-Enclosed Switchgear and Control gear”

IEC 60056 “HV Alternating-Current Circuit Breakers”

IEC 60815 “Selection and dimensioning of high-voltage insulators intended for use in polluted conditions”

IEC 60273 “Characteristic of indoor and outdoor post insulators for systems with nominal voltages greater than 1000V”

IEC 60168 “Tests on indoor and outdoor post insulators of ceramic material or glass for systems with nominal voltages greater than 1000 V”

IEC 60282 “HV Fuses”

IEC 60529 “Classification of Degrees of Protection Provided by Enclosures”

IEC 60947 “LV Switchgear and Control gear”

IEC 60158 “LV Control gear”

IEC 61439 “LV Switchgear and Control gear assemblies”

IEC 60408 “LV Air-break Switches, Air-break Disconnectors, etc.”

IEC 60269 “LV Fuses”

IEC 62040 “Uninterruptible power systems (UPS)”

Battery

IEC 61056 “General Purpose Lead-Acid Batteries ”

IEC 60623 “Secondary Cells and Batteries containing Alkaline or other Non-Acid Electrolytes – Vented Nickel Cadmium Prismatic Rechargeable Single Cells

IEC 60896 “Stationary Lead-Acid Batteries”

Capacitor

IEC 60871 “Shunt Capacitors for A.C. Power Systems having a rated voltage above 1000V”

Metering

IEC 62053 “Electricity Metering Equipment (A.C.)”

Communication

IEC 61850 “Communication Networks and Systems for Power Utility Automation” (for substation power utility automation IEDs)

Cables

IEC 60230 “Impulse tests on cables and their accessories”

IEC 60228 “Conductor of insulated cables”

IEC 60229 “Tests on cables over sheath which have a special protective function and are applied by extrusion”

IEC 60840 “Extruded solid dielectric power cables for rated voltages above 30 kV”

IEC 60811 “Common tests methods for insulating and sheeting materials of electric cables”

IEC 61238 “Compression and mechanical connectors for power cables with copper or aluminium conductors”

IEC 60287 “Electric cables – calculation of the current rating”

IEC 60502 “Extruded solid dielectric power cables for rated voltages from 1 kV up to 30 kV”

IEC 60304-1 “Standard Colours for Insulation of low Frequency Cables and Wires”

IEC 332-2 Cat. C Tests on electrical cables under fire conditions

ANSI/IEEE 386 “Standard for separable Insulated Connector Systems for Power Distribution Systems Rated 2.5kV through 35 kV”

Electromagnetic compatibility (EMC)

IEC 6100-4/255-6 "Electromagnetic compatibility (EMC) - Testing and measurements Techniques

Protection of apparatus

IEC 60137 “Bushings for alternating voltages above 1000 V”

IEC 60233 “Test on hollow insulators for use in electrical equipment”

IEC 60722 “Guide to the lightning impulse and switching impulse testing”

IEC 60099 “Surge arresters”

IEC 60358 “Surge Capacitor”

IEC 62305 “Lightning Protection Standard”

IEEE 80 “Guide for Safety in AC substation grounding”

IEC 60079-10 “Electrical apparatus for explosive gas atmospheres Part 10: Classification of hazardous areas”

IEC 60079-14 “Explosive atmospheres – Part 14: Electrical installations design, selection and erection”

IEC 60255 “Measuring Relays and Protection Equipment”

IEC 61010 “Safety requirements for Electrical Lab Equipments”

IEC 60529 “Degrees of protection provided by enclosures (IP Code)

ANSI C3-.23 “Standard for Metal-Enclosed Bus”

Installation

IEC 60364 “Low Voltage Electrical Installations”

IEC 60423 “Conduit Systems for Cable Management”

IEC 60669 “Switches for household and similar fixed-electrical installations”

Lighting

IEC 60081 “Double-capped fluorescent lamps”

IEC 60118 “High – pressure Mercury Vapour Lamps”

IEC 60192 “Low – pressure Sodium Vapour Lamps”

IEC 60662 “High – pressure Sodium Vapour Lamps”

IEC 60357 “Tungsten Halogen Lamps”

Control & Instrumentation

ISA “International Society of Automation”

ISA S51.1 “Process Instrumentation Terminology”

ANSI FCI 70-2 “Valve seat leakage class”

ISO 5167 “Measurement of fluid flow by means of pressure differential devices inserted in circular cross-section conduits running full”

ASME PTC 19.5 “Flow measurement”

IEC 144 :1963 “Degrees of Protection of Enclosures of Switchgear and Control Gear for voltages up to and including 1000 VAC and 1200 VDC, BS 5420”

ISA RP 55.1 “Hardware Testing of Digital Process Computers”

IEC 60751 “Industrial platinum resistance thermometers and platinum temperature sensors”

IEC 60584 “Thermocouples”

IEC 61508/61511 “Functional safety of electrical/electronic/programmable electronic safety-related systems

Fire Fighting

NFPA 10 "Portable fire extinguisher"

NFPA 11 “Standard for Low-Expansion Foam”

NFPA 2001 “Standard on Clean Agent Fire Extinguishing Systems”

NFPA 12 "Carbon dioxide extinguisher system"

NFPA 13: Standard for the Installation of Sprinkler Systems

NFPA 14 “Installation of standpipe, private hydrant s and hose system”

NFPA 15 "Water spray fixed systems for fire protection"

NFPA 16 “Foam extinguishing systems”

NFPA 20 “Installation of stationary pumps for fire protection”

NFPA 24 - Standard for the Installation of Private Fire Service Mains and their Appurtenances

NFPA 30 "Flammable and combustible liquid code"

NFPA 70 “National Electrical Code”

NFPA 72 "National fire alarm code"

NFPA 850 "Fire protection for fossil fuelled steam and combustion turbine electric generating plants"

ISO 834 “Fire Resistance Tests - Elements of Building Construction”

API RP 2030 “Application of Fixed Water Spray Systems for Fire Protection in the Petroleum and Petrochemical Industries”

Noise Emission

ISO 1996-2:2007 “Measurement and assessment of environmental noise”

ISO 3746:1995 “Determination of sound power levels of noise sources using sound pressure

Survey method using enveloping measurement surface over a reflecting plane”

I.S. 5739 “Unreasonable noise for building equipment”

Coding

KKS coding shall be applied

Performance Tests

ASME PTC 22 - 2005 “Performance Test Code on Gas Turbines

ASME PTC 46 – 1996 “Performance Test Code on Overall Plant performance”

ASME PTC 4.4 "Gas turbine Heat Recovery Steam Generator"

ASME PTC 6.2 "Steam Turbine"

ASME PTC 4.0 " Fired Steam Generators"

VGB-R 131M Air Cooled Condensers under vacuum Performance test

F

ASME PTC10 "Compressors and exhausters"

ASME PTC 19.11 – Steam and Water Sampling, Conditioning, and Analysis in the Power Cycle - Performance Test Codes

ASME PTC 30.1 “Air Cooled Condensers”

ISO 2314 Gas Turbines – Acceptance tests

DIN 1942 Heat Recovery Steam Generator Performance Tests

DIN 1943 Thermal acceptance tests of Steam turbines

ISO 5198 Centrıfugal, Mixed Flow and Axial Pumps. Code for Hydraulic Performance Tests

HIS Hydraulic Institute Standard

VDI :2056 Norms and Procedures for Vibration Assessment on Equipment

NS 05-0062 - Air Pollution -Discharge Standards

3.2. UNITS AND LANGUAGE REQUIREMENTS

3.2.1 Units of Measurement

SI units shall be used in all correspondence, documentation, calculations, drawings, measurements etc. For temperature, pressure, flow data, to use the metric units “°C” “bar” and “kg/s” or “t/h” respectively. If reference has to be made to non-standard items, other than for temperature (°C) and pressure (bar), the SI units shall be quoted followed by the non-standard units in brackets.

General arrangement drawings and system arrangements that are produced specifically for the project shall be produced to metric dimensions.

“All equipment used in the plant for measurement and control equipment will be calibrated in the International System (SI) units of measurement adopted by the International Bureau of Weights and Measures.”

3.2.2 Language Requirements

French is the sole language of the contract between EMPLOYER and CONTRACTOR. So, the documents shall be submitted as both English and French at the execution stage by SUPPLIER to CONTRACTOR.

(English Document – CONTRACTOR’s Review, French & English Document – Submission document to EMPLOYER)

Attachment-A : Natural gas analysis

	MDT samples			Tortue-1 DST
	Tortue-1	Ahmeyim-2	Guembeul-1A	LC-B BHS1.76 CVD
TVDss (m)	4352.71	5025.9	4627.4	
Sands	096_Lower Cenomanian	106_Albian	106_Albian	LC-B 30%FlowRate BHS
Sample No.	2.02	2.03	2.07	1.76
N2	0.2177	0.1583	0.1287	0.1553
CO2	0.3265	0.6529	0.5443	0.3285
CH4	92.7748	92.6346	93.3717	93.1802
C2H6	3.2259	3.2850	2.9990	3.1928
C3H8	1.2783	1.2980	1.1396	1.2788
IC4	0.2295	0.2096	0.1693	0.2193
NC4	0.3828	0.3334	0.2911	0.3753
IC5	0.1390	0.1096	0.0854	0.1426
NC5	0.1226	0.0935	0.0788	0.1221
C6	0.1427	0.1244	0.0957	0.1363
Benzene	0.0945	0.0775	0.0810	0.0100
Toluene	0.0134	0.0646	0.0753	0.0379
EthBenz	0.0155	0.0143	0.0150	0.0045
Xylene	0.0463	0.0504	0.0591	0.0333
C7	0.1915	0.1353	0.1131	0.2114
C8	0.2161	0.1731	0.1722	0.1853
C9	0.0933	0.0638	0.0624	0.0709
C10	0.0952	0.0836	0.0857	0.0767
C11 - C14	0.1941	0.1718	0.1746	0.1382
C15 - C29	0.1393	0.1242	0.1371	0.0934
C30+DST	0.0000	0.0000	0.0000	0.0070
C30+MDT	0.0610	0.1421	0.1209	0.0000
Sum	100.0	100.0	100.00	100.00

Attachment-B : Naphtha Analysis

Société Africaine de Raffinage
Direction Audit et Conformité / Département Laboratoire

Bulletin Analyse N°288573

Produit : NAPHTA **Reservoir : T511-T512-T514-T515**

Bon de Commande :
Date de Reception : 2020-01-21
Heure de Reception : 11:00:00
Validation faite par le matricule : 636
Analyse Le : 2020-01-21 18:16:53
Destinataire : EXPORT

Echantillon : 6784

METHODE	CARACTERISTIQUE	UNITE	RESULTATS
ASTM D5191	Pression de vapeur ? 37.8 ?C	g/cm2	120
ASTM D6733	AROMATIQUE (% v/v	% (v/v)	14.4
ASTM D1266	Teneur en Soufre UV	% m/m	0.01
ASTM D86	Distillation - Analyse Point Initial	°C	85.4
ASTM D86	Distillation Analyse 05%	°C	107.4
ASTM D86	Distillation Analyse 10%	°C	112.7
ASTM D86	Distillation - Analyse 20%	°C	118.6
ASTM D86	Distillation - Analyse 30%	°C	123.4
ASTM D86	Distillation - Analyse 40%	°C	128.3
ASTM D86	Distillation - Analyse 50%	°C	133.2
ASTM D86	Distillation - Analyse 60%	°C	138.2
ASTM D86	Distillation - Analyse 70%	°C	143.9
ASTM D86	Distillation - Analyse 80%	°C	151.8
ASTM D86	Distillation - Analyse 90%	°C	160.4
ASTM D86	Distillation - Analyse 95%	°C	166.7
ASTM D86	Distillation - Analyse Point Final	°C	178.8
ASTM D86	Distillation - Analyse Residu	% v/v	1.0
ASTM D86	Loss/Perte	% vol	0.6
ISO 12185	Masse Volumique à 15°C	Kg/m3	767.2
ASTM D2699	Indice Octane (NOR)		53.3
NF EN ISO 12937	Teneur en eau	(% m/m)	0.01

POUYE BABACAR
Société Africaine de Raffinage Service Laboratoire
CHIMISTE RESPONSABLE DE QUART

Page 1/1

Km 18 Route de Rufisque - Téléphone (00221)338398447 - Email: sar@sar.sn

FOR 216

Pour déclarer ou non la conformité, il n a pas été tenu compte de l incertitude associée au résultat.

Les résultats ne se rapportent qu aux objets soumis à essais

Attachment-C : Gas Oil analysis

AFRICAN REFINERY COMPANY
DIRECTORY OF TECHNICAL/SERVICE LABORATORY

ANALYSIS REPORT NO. 264996
Product: LIGHT GAS Reservoir: T511 T 513 T516

Purchase Order: 1
Date of Reception: 2018-12-19 Sample Number: 3249
Time of Reception: 10:00:00
Validation made by Personnel: 636
Analysis: 2018-12-20 09:54:57
Recipient: African Refinery Company

Method	Characteristics	Unit	Results
D1319	Aromatic	Volume %	5.35
ASTM D1266	Sulfure Content UV	% m/m	0.0001
ASTM D86	Distillation - Initial Point Analysis	C	37
ASTM D86	Distillation Analysis - 5%	C	52.1
ASTM D86	Distillation Analysis - 10%	C	55.1
ASTM D86	Distillation Analysis - 20%	C	59.7
ASTM D86	Distillation Analysis - 30%	C	63.9
ASTM D86	Distillation Analysis - 40%	C	68.2
ASTM D86	Distillation Analysis - 50%	C	72.7
ASTM D86	Distillation Analysis - 60%	C	77.35
ASTM D86	Distillation Analysis - 70%	C	82.0
ASTM D86	Distillation Analysis - 80%	C	87.5
ASTM D86	Distillation Analysis - 90%	C	95.7
ASTM D86	Distillation Analysis - 95%	C	103.3
ASTM D86	Distillation - Final Point Analysis	C	127.8
ASTM D86	Distillation Residual Analysis	% v/v	1.1
ISO 12185	Volumic Mass at 15C	kg/m3	702.6
ASTM D2699	Octane Number		72.4
NF M07016	Steam (Vapor) Pressure 37.8C	g/cm2	581
NF EN ISO 12937	Water Content	% m/m	0.01

Signed by
POUYE BABACAR

CHEMIST RESPONSIBLE FOR THE TEST

Attachment-D : Raw Water Analysis

USF PERRIER

3

DATA SHEET

2- DONNEES DE BASE

2-1. Qualité de l'eau brute

Les paramètres sont ceux ayant servi de base au présent projet.

Dans le cas où certains paramètres ne seraient pas conformes à ceux indiqués, les capacités, les résultats et les consommations de réactifs pourraient en être affectés.

		Données SENELEC	Base de dimensionnement
CATIONS	:		
Calcium	:	3,84 méq/l	3,84 méq/l
Magnésium	:	0,96 méq/l	0,96 méq/l
Sodium	:	3,64 méq/l	5,12 méq/l
Potassium	:	0,46 méq/l	0,46 méq/l
Baryum	:	NC	0,00 méq/l
Strontium	:	NC	0,00 méq/l
Total Cations	:	8,90 méq/l	10,38 méq/l
ANIONS	:		
TAC	:	6,24 méq/l	6,24 méq/l
Chlorures	:	3,60 méq/l	3,60 méq/l
Sulfates	:	0,48 méq/l	0,48 méq/l
Nitrates	:	0,06 méq/l	0,06 méq/l
Fluorures	:	NC	0,00 méq/l
Total Anions	:	10,38 méq/l	10,38 méq/l
Fer	:	1,1 mg/l	1,1 mg/l
Matières organiques	:	0,1 à 3 mg/l	0,1 à 3 mg/l
Turbidité	:	1 à 10 NTU	1 à 10 NTU
Conductivité	:	520 à 1000 µS/cm	520 à 1000 µS/cm
pH	:	6,5 à 8	7
Température	:	22 à 29°C	25°C
Silice SiO_2	:	36,0 mg/l	36,0 mg/l maxi

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